Portfolio Specification

Data Science & Machine Learning 1 – Software Development Year 4



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1. Overview and Background

In Semester 1 of Year 3 we were taught Python, in parallel with coursework I built a Last.Fm scrobbling application that connects with the Last.Fm API to track my music listening in order to step away from Spotify and subscription streaming services. I deployed the application to pythonanywhere, as shown in our Python Web Dev classes.

A year has passed, and as such I now have a year's worth of data that would allow me to personally connect with my portfolio. Figure 1 shows just under 18,000 entries.



Figure 1: 17,927 data points tracking my music listening over the last 365 days

1.1 Dataset

- **Source 1:** Data can be pulled from the Last.Fm API. The endpoint of focus for this portfolio is `user.getRecentTracks`, which returns a users own scrobble history and `user.getLovedTracks`, which will be joined onto the scrobble data as a binary field
- **Source 2:** Hugging Face Spotify dataset, a 45 million track dataset. This paraquet files are loaded and filtered in chunks using `pandas.read_paraquet()`. Then merged with the scrobble dataset for more features
- **Size:** ~18,000 timestamped scrobbles collected personally via API key, ~1800 unique merged tracks
- **Fields:** `artist, album, song, timestamp, is_loved` | Additional features like `hour, tod` and `release_year`

The raw Last.Fm data is returned as a nested JSON with an embedded #text key for artist, album and date fields. This is detailed in the Unofficial Last.Fm API docs [2]. The first step of preparing the data is extracting these fields into a readable table.

The portfolio analyses my own listening data. No other users' data is collected, and API credentials are stored in a local, git ignored `lastfm_creds.py` file

2. Technologies

- **Language:** Python
- **Data Science & ML libraries:** pandas, scikit-learn
- **Visualisation:** matplotlib
- **Environment:** Jupyter Notebook
- **API Access:** requests module
- **External Tools:** Last.Fm API, Hugging Face Hub
- **Version Control:** GitHub

3. Questions to Explore

3.1 Linear Regression – Release Year Trend

Does the average release year of the music I scrobble change from week to week?

- Group scrobbles by week and get the average release year per week
- Filter out weeks with less than 25 plays
- Fit a Linear Regression model with time as the predictor

Listeners usually listen to older music over a longer period of time as they revisit old favourites. Positive slop means I drift towards newer releases

3.2 Linear Regression – Daily Scrobble Count

How many songs Do I scrobble per day, and what external factors affect it?

- Follow the structure of the `PythonDataScienceHandbook` bicycle example
- Use seven day of week indicator
- Note Irish public holidays and hours of daylight for Dublin's latitude (53.35 N)

Test whether my listening follows similar patterns to the bicycle trips by implementing the calendar driven method from the handbook

3.3 Logistic Regression – Predicting Artist from Time of Day

Can the time of day help with predicting which of my ten most listened to artists I am likely to listen to?

- Multiclass Logistic Regression model trained on one-hot encoded time of day features (morning, afternoon, evening, night)
- Compare against a baseline that predicts the most common artist
- Binary Logistic Regression Model for a top artist
- Predict the probability that a scrobble is that artist given the hour of day

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References

[1] Last.Fm (n. d.) *User.getRecentTracks* [API documentation] Available at: https://www.last.fm/api/show/user.getRecentTracks (Accessed: 05th October)

[2] Last.Fm API Docs (n.d.) *User.getRecentTracks* [Unoficcial Last.Fm API docs] Available At: https://lastfm-docs.github.io/api-docs/user/getRecentTracks/ (Accessed: 05th October 2026)

[3] Carlos Medina (2021) *Using the Last Fm API to extract data and analyzing it with Python* [Hemisferios] Available At: https://medium.com/hemisferios/using-the-last-fm-api-to-extract-data-and-analyzing-it-with-python-part-1-ffe96f812455 (Accessed: 05th October 2026)

[4] Mark Lambert (2012) *Personal LastFm Profile – Source of Scrobbling data* [LastFm] Available at: https://www.last.fm/user/MilesOver666